Assessment Task - 1

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CSA0519 - Database Management System

AT - 1

(1) Vehicle Management System - EER Diagram

(a) Identify the Supertype and Subtypes

Supertype: Vehicle

Subtype: Car and Truck

Vehicle is the Supertype that contains common attributes. Car and truck are specialized entities that inherit the common attributes of vehicle through an ISA relationship.

(b) Common attributes of Vehicles:-

1. Vehicle - ID → primary key
2. Registration - Number
3. Manufacturer
4. Model
5. year.
6. Purchase - price

Specialized Attributes of Car and Truck

Car Attributes:-

No: of Door

No: of fuel type

Seating - Capacity.

Truck Attributes

Load - Capacity

Truck - type

No. of Axles.

④ primary key of Inheritance

Vehicle-ID is the primary key of vehicle supertype. Both Car and truck inherit this primary key along with the common attributes of Vehicle.

⑤ ISA Specialization

⇒ Car inherit all common attributes from vehicle

⇒ Truck inherit all attributes from vehicle

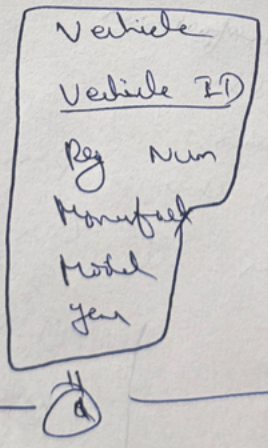
⑥ Disjoint or Developping

⇒ Vehicle Car belong to either Car or truck

Subtypes Disjoint is

⑦ Total or partial

The specialization is total



Vehicle-ID

Fuel Type Seating Capacity

Truck

Vehicle Id

Load Capacity

No. of Axles.

Bank Management System:

① The major entities are Customers, Account, Branch, Employee, Transaction, Loan, Loan-payment.

② & ③

Attributes and primary keys

Customers: Customer-ID, Customer name, Address, phone, Email.

Account: Account-NO

Branch: Branch-ID

Employee: Employee-ID

Transaction: Transaction: ID

Loan: Loan-ID

Loan-Payment: Payment-ID:

④ & ⑤

- Customer-Account 1:N; one customer may hold multiple accounts.
- Branch-Account.
- Account Transaction
- Customer Loan
- Loan Segment.

ER Diagram:

Customer

Customer - ID

Customer - Name

Address

Loan

Loaded - LD

Loan - Type

Interest Rate

Account

Acount - NO.

Acount - Type

Balance

Transaction

Transaction ID

Date

Type

Branch

Branch - ID

Branch - Name

Lister

Employee

ID

Name

Salary

E - Commerce website

① Major Entities.

* Customer, products, Category, Order, payment and delivery

② Attributes at primary key

- Customer : Customer - ID
- product : product - ID
- Category : Category - ID
- Delivery : Delivery - ID

③ Order - product Relationship

* Order and product have my to many relationship.

↳ attributes qunty and unit price are stored in order - item.

④ product is Supertype.

Subtypes are

⇒ Electronics.

⇒ clothing

⇒ Grocery.

6. Specialization Constraints.

- Disjoint → only belongs these Subtypes
- Total → Every product belongs to these Subtypes.

7.

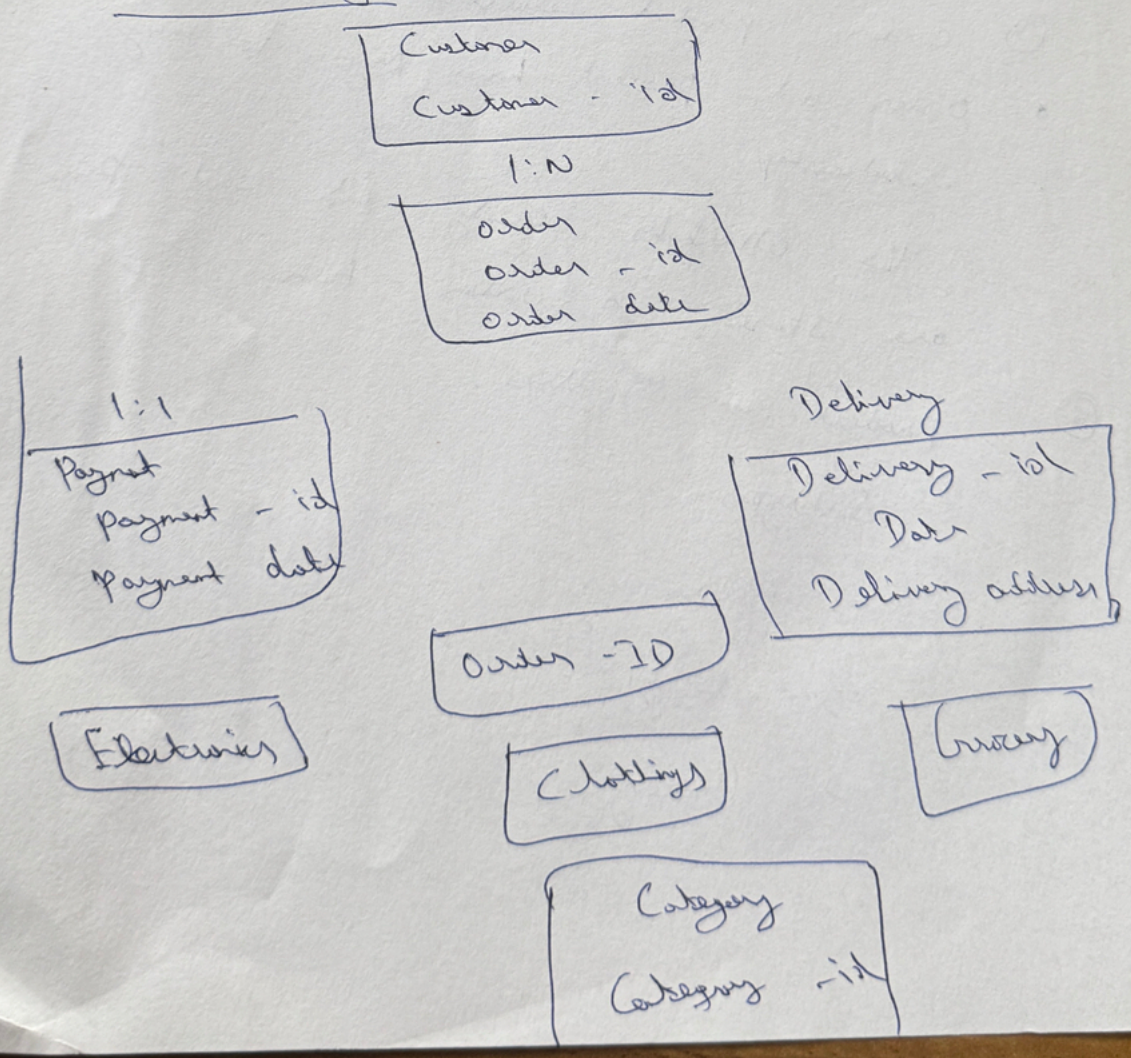
8. Relationship and Cardinalities.

Customer - order → 1:N

order - payment → 1:1

order - Delivery → 1:1

(9) EER Diagram



Hospital Management System

1. Major Entities

The major entities are Staffs, Doctor, Nurse, Admin, Department, Appointment.

2-4 Staffs Specialisation and attributes.

Staffs	Doctor
Staffs-id	• Staffs-id
Name	• Specialisation
Address	• Licence-Number.
phone	
Salary	

8-9

Department - Relationship

Dept - Doctor	1:N
patient - Appoin	1:N
Doctor - Appin	1:N
patient - Medical - Record	1:N

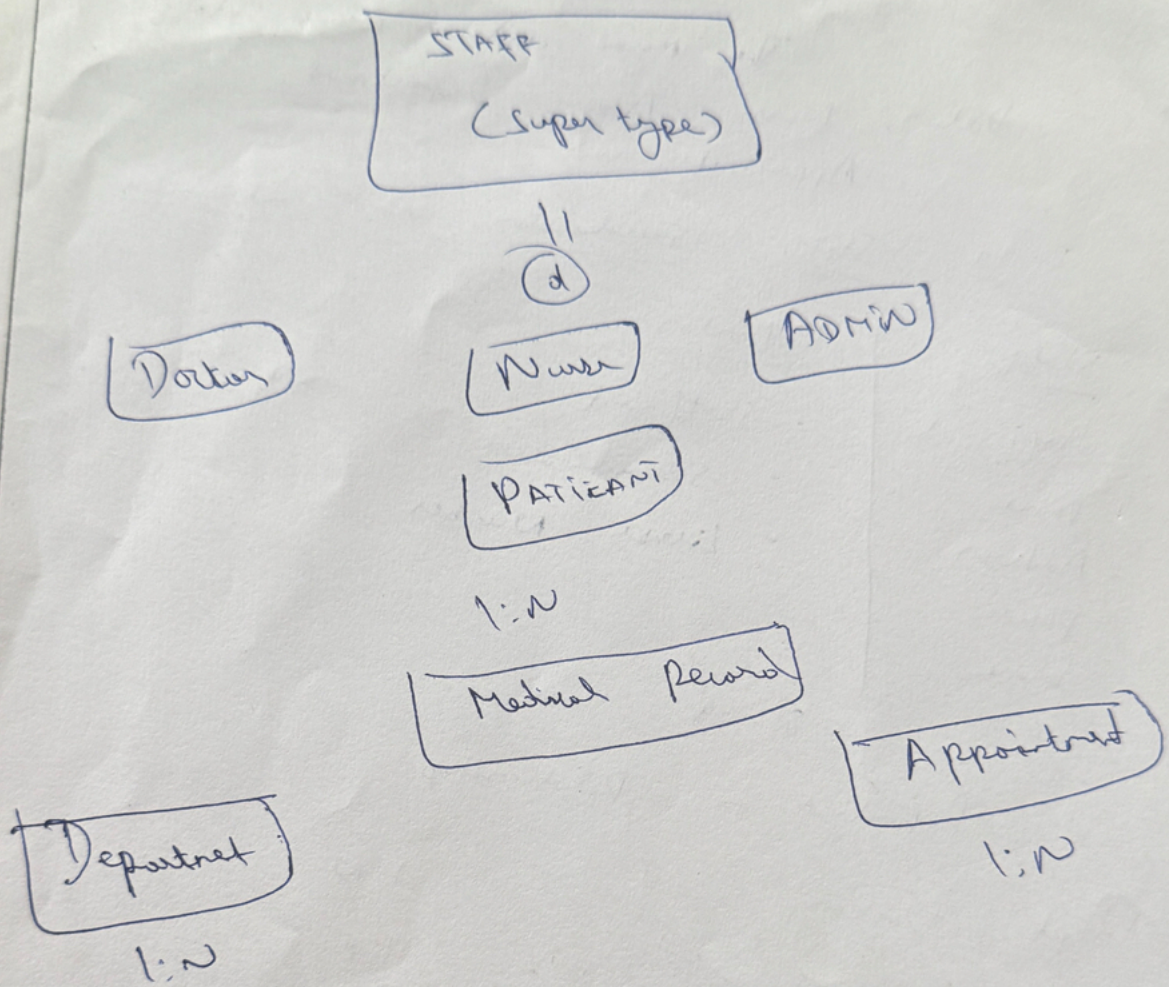
5-7 Week entity and Identifying

Medical record is week entity.

partial key - Record Rule.

Identifying relationship - HAS - Record.

EER Diagram


STAFF
(Super type)
d
Doctor
Nurse
ADMIN
PATIENT
1:N
Medical Record
Appointment
Department
1:N
1:N